Problem Set 10

Due date: 12/5/2025

Note: the problems are organized in the order in which we cover the material (some problems will be on material covered the lecture after Thanksgiving). Problems marked SR are to be done using only special relativity – i.e. ignoring the gravitational field.

1. *(10 points)* (SR) Derive the Lorentz transformations for $\vec{E}$ and $\vec{B}$ under a boost in the $+x$ direction.

2. *(10 points)* (SR) Write down the components of $*F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\lambda\rho}F_{\lambda\rho}$.

3. *(15 points)* The spacetime diagram below is drawn in the reference frame of a small lab sitting on the earth (which experiences a Newtonian gravitational field of $g = 10m/s^2$). Which of the worldlines shown below has the greatest proper time (note that different curves may have differential initial velocities)? Explain why the answer you chose is correct.

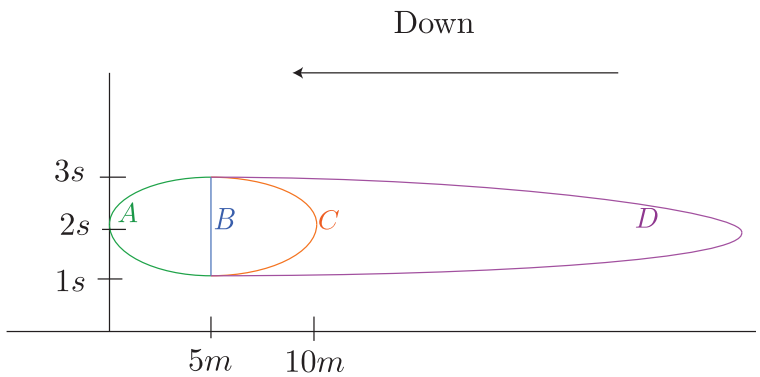

Figure 1: Diagram for problem 4. Each letter refers to the worldline of the same color.

4. *(10 points)* Use the mass and radius of the earth to calculate how much faster a clock in Denver runs than a clock in Washington, D.C. does. Denver is 1600m higher than Washington, D.C.

5. *(25 points)* Consider a new parametrization of a geodesic: instead of τ, we will parametrize it with some new parameter σ.

   (a) *(20 points)* Show that the geodesic equation in terms of such an arbitrary parameter can be written:
   $$\frac{d^2x^\beta}{d\sigma^2} + \Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} = \frac{d}{d\sigma}\ln\left(\frac{d\tau}{d\sigma}\right)\frac{dx^\beta}{d\sigma}$$

   (b) *(5 points)* Show that if $\sigma = a\tau + b$ for some constants a and b, we get back the usual geodesic equation we derived in class.

6. *(15 points)* Prove that $U^aU_a = -1$ for timelike geodesics.

7. *(50 points)* Consider the following metric:
   $$ds^2 = -\left(1 + \frac{r^2}{\ell^2}\right)dt^2 + \frac{dr^2}{1 + \frac{r^2}{\ell^2}} + r^2 d\Omega^2.$$

where ℓ is some fixed length.

(a) *(15 points)* Use the fact that every null curve in two dimensions is a null geodesic to solve for the trajectory of a radial light ray (a light ray that travels along paths of constant θ and ϕ).

(b) *(10 points)* Suppose you are a stationary observer at $r = 0$, and you turn on a flashlight at $t = 0$. At what value of t would the radial light rays from your flashlight reach $r = \infty$? Call this time t_∞. What is the value of r at $2t_\infty$?

(c) *(10 points)* What do you see at your original position $r = 0$, $t = 2t_\infty$?

(d) *(15 points)* This metric does not depend on t. It turns out that this means that the following quantity is constant along timelike geodesics:

$$-\left(1 + \frac{r^2}{\ell^2}\right)\frac{dt}{d\tau},$$

where τ is a proper time. Call this quantity C. Use this and the fact that $U_a U^a = -1$ for a timelike geodesic to show that timelike geodesics cannot reach $r = \infty$ but instead have a maximum at some finite value of r. [hint: $dr/d\tau = 0$ at a maximum]. Find the value of r at the maximum.

Note: this means that timelike geodesics move like oscillating boomerangs, constrained to stay away from $r = \infty$! This spacetime is often said to be like "gravity in a box": gravity compels observers to stay near $r = 0$.

8. *(15 points)* Timelike geodesics maximize proper time while null geodesics have zero elapsed proper time. Do spacelike geodesics maximize proper distance? Minimize proper distance? Neither?